

AI-Powered Skin Disease Detection: a Deep Learning Approach with ResNet50 and EfficientNetB0

Presented by

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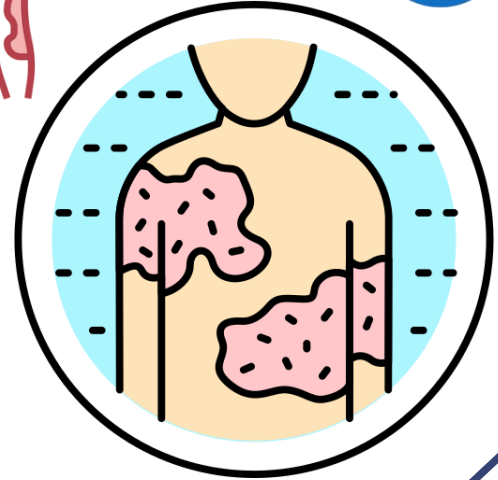
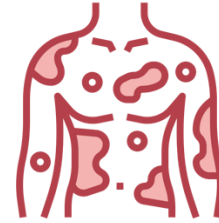
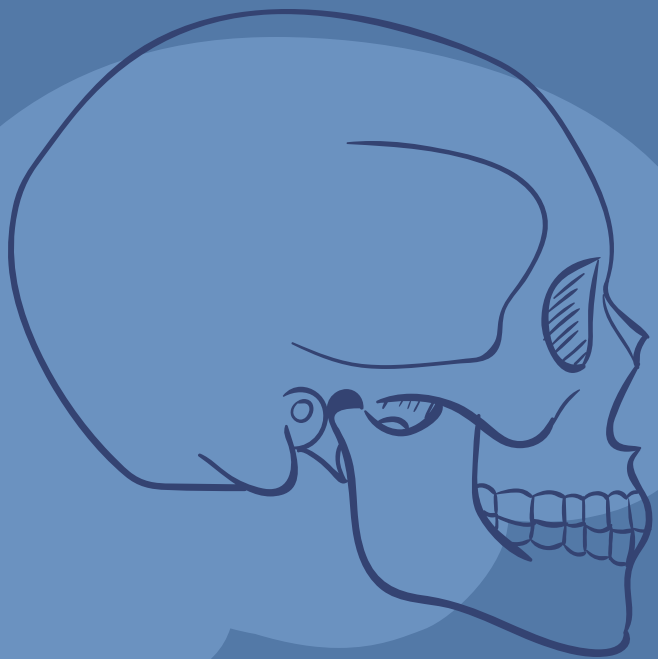


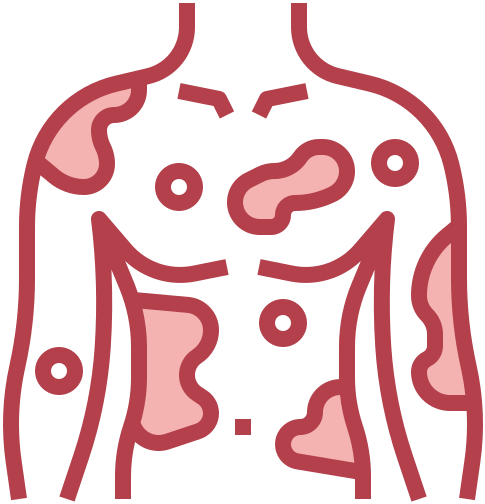
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INTRODUCTION



WHAT IS Skin Disease?

A skin disease or disorder is a condition that affects the skin's **structure, function, or appearance**, leading to irritation, infection, or abnormalities.

- Impact: Can cause **itching, redness, pain, blisters, or pigmentation changes**, affecting both **health and quality of life**.
- Common types are: eczema, psoriasis, melanoma etc.

PROBLEM STATEMENT

Limited access to dermatologists delays skin disease diagnosis. This study proposes an AI-powered detection system using deep learning for fast and accurate classification. A web-based platform enhances accessibility for early intervention.





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RELATED WORKS

Related Work

Relevant Studies:



- Deep learning models (ResNet, InceptionNet, DenseNet) improve skin disease classification.
- Hybrid models (CNN + Decision Trees/LSTMs) enhance precision.

Limitations



- Class imbalance affects rare disease detection.
- Limited dataset diversity reduces generalization.
- Lack of explainability hinders clinical adoption.

Gaps Addressed



- Optimized ResNet50 & EfficientNetB0 for better accuracy.
- Data augmentation to handle class imbalance.
- Web-based tool for real-time, accessible diagnosis.

Research Questions

•How effectively can deep learning models classify various skin diseases using ResNet50 and EfficientNetB0?



•What impact does data augmentation have on improving classification accuracy and addressing class imbalance?



•How can a web-based AI system enhance accessibility and early diagnosis of skin diseases?

•What are the key challenges in deploying AI-powered skin disease detection in real-world clinical settings?



OBJECTIVE

- Develop an AI-powered skin disease detection system using **ResNet50** and **EfficientNetB0**.
- Improve classification accuracy through **data augmentation** and **preprocessing techniques**.
- Build a **web-based platform** for real-time and accessible diagnosis.
- Enhance model generalizability and reliability for **real-world clinical applications**.

Outcomes and Impacts

Outcomes (Points)

Enhanced Classification Accuracy – Improved skin disease detection with ResNet50 & EfficientNetB0.

Robust Model Performance – Addressed class imbalance using data augmentation techniques.

Web-Based Diagnosis System – Developed a real-time, accessible platform for users.

Improved Generalization – Model trained on diverse datasets for better real-world applicability.

Optimized Processing – Efficient deep learning model suitable for deployment in healthcare.

Impacts (Checklist)

- ✅ **Early Diagnosis** – Enables faster identification of skin diseases, leading to timely treatment.
- ✅ **Increased Accessibility** – Provides an AI-driven tool for individuals without easy access to dermatologists.
- ✅ **Reduced Healthcare Costs** – Minimizes unnecessary doctor visits for common skin conditions.
- ✅ **Scalability for Global Use** – Can be extended to different demographics and healthcare systems.
- ✅ **Potential for Clinical Adoption** – Supports dermatologists in decision-making and diagnosis.

METHODOLOGY

Data Collection & Preprocessing: Collected **20,306 images** from public dermatology datasets (ISIC Archive, DermaMNIST). And Image resizing (224×224), normalization, label encoding, and augmentation (rotation, zoom, flip, shift).

FlexiGrid System: Designed a **scalable and adaptive framework** for efficient image processing and classification. Also Ensured **diagnosis** by optimizing computation and resource allocation.

Models & Techniques:

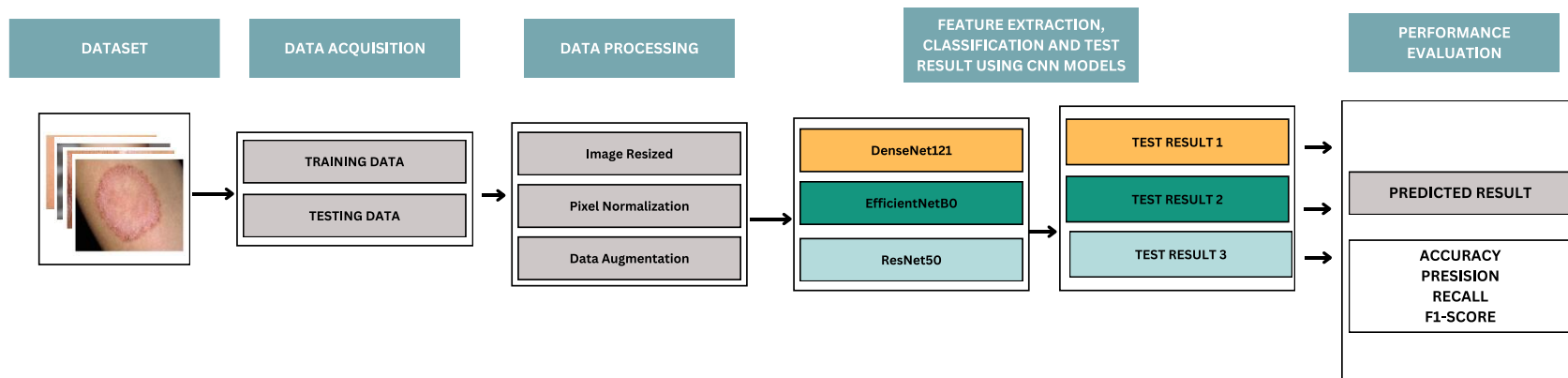
- **Deep Learning Models:** Implemented **ResNet50** and **EfficientNetB0** for feature extraction.
- **Transfer Learning:** Used pre-trained models for improved accuracy.
- **Evaluation Metrics:** Accuracy, precision, recall, F1-score, and confusion matrix..

DATASET DESCRIPTION



- Collected from: Kaggle, DermaMNIST , ISIC Archive
- Number of Classes: Four (16)
- Name of the Classes: Glioma, Meningioma, Pituitary, and No Tumour
- Total Samples: 20,306 2D image

WORKFLOW



Results & Key Findings

Results

The study successfully implemented **ResNet50** and **EfficientNetB0** for skin disease classification, achieving **validation accuracies of 83.61% and 90%, respectively**. Data augmentation and preprocessing techniques significantly enhanced model robustness. The **web-based platform** provided **real-time diagnosis**, making AI-driven dermatology **more accessible and efficient**. The use of **learning rate optimization and early stopping** improved training stability, preventing overfitting.

Findings

- **EfficientNetB0 outperformed ResNet50**, demonstrating superior classification accuracy.
- **Data preprocessing (augmentation, normalization) enhanced model performance** and generalization.
- **Class imbalance remains a challenge**, requiring more diverse datasets for better real-world applicability.
- **The web-based system enables scalable, real-time skin disease detection**, improving accessibility to dermatological care.

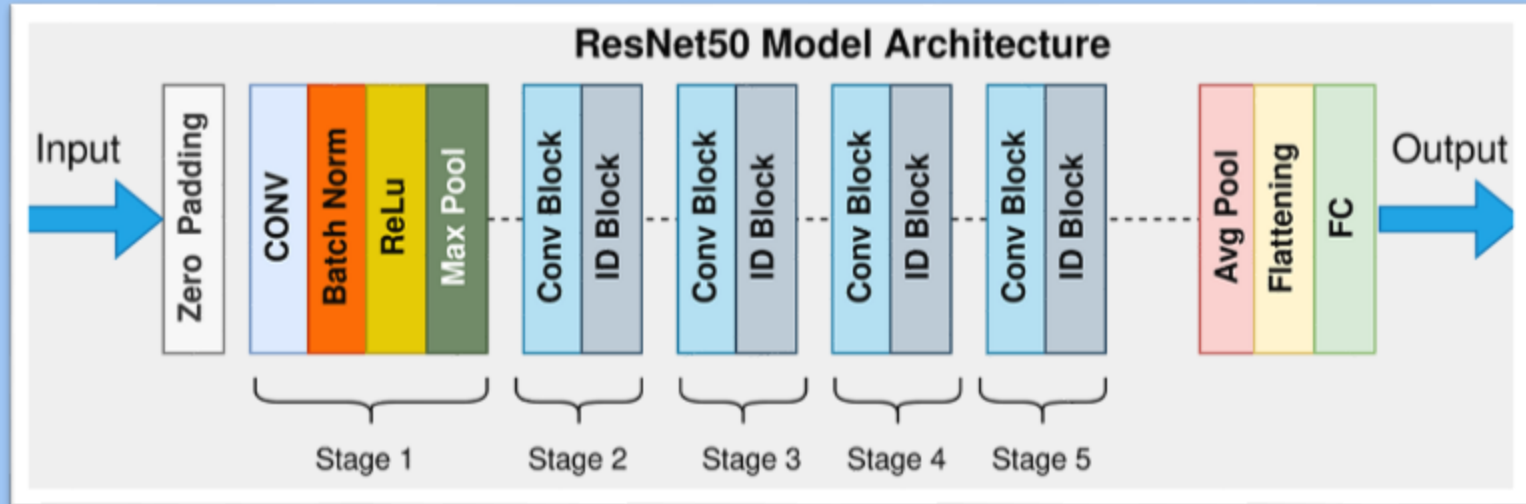
DATA PREPROCESSING

- Sample images were resized to a fixed $224 \times 224 \times 3$ pixels
- Min-max normalization was applied to the resized photos to scale the intensity values between 0 and 1
- Data augmentation techniques such as horizontal flip, rotation zoom, height shift, and width shift were applied



MODEL Description

ResNet50: ResNet50 architecture employs 50 layers with residual connections, allowing information to bypass certain layers, which helps prevent the vanishing gradient problem and enables training of much deeper networks.

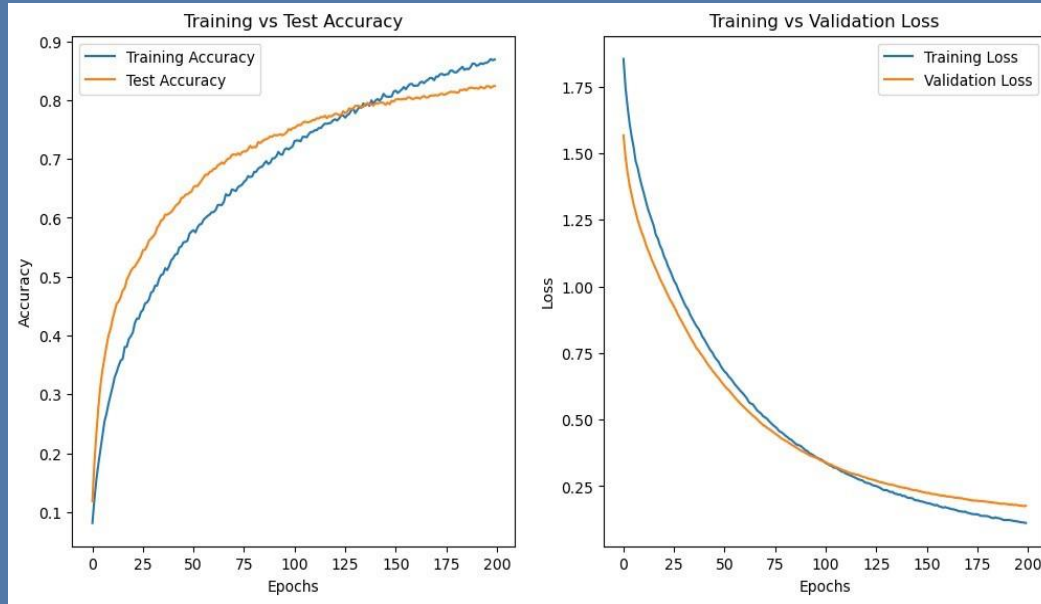


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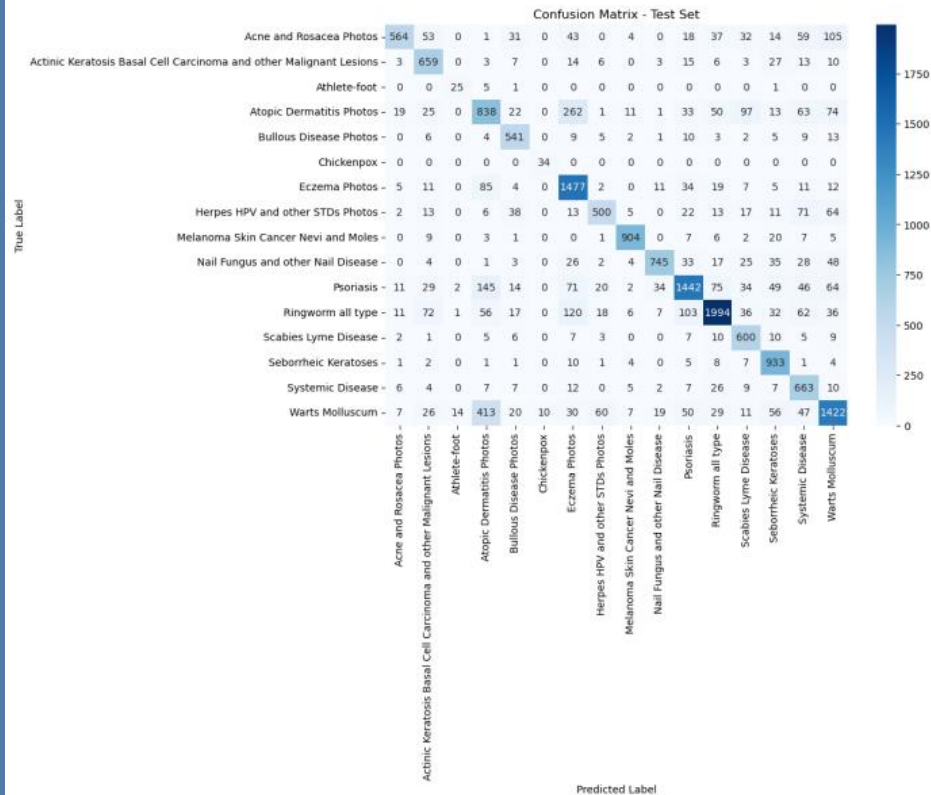


RESULT ANALYSIS

TRAINING, TEST & VALIDATION ACCURACIES (EPOCH VS. ACCURACY)



CONFUSION MATRIX (ResNet50)



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CONCLUSION

COMPARATIVE STUDY

This study developed an **AI-powered skin disease detection system**, selecting **ResNet50** for its **real-world performance, stability, and adaptability** in healthcare applications. While **EfficientNetB0** achieved **higher validation accuracy (90%)**, **ResNet50** proved **optimal for practical deployment** due to its robustness and efficiency.

RESULT ANALYSIS

The **web-based platform** enables **real-time, cost-effective skin disease detection**, improving accessibility. Future work should **expand dataset diversity, enhance explainability, and optimize for telemedicine**. This research demonstrates **ResNet50's practicality**, offering a **scalable AI-driven solution** for **early dermatological diagnosis**.

Future Research Direction

- **Expand Dataset Diversity** – Improve generalization across skin tones and conditions. Enhance Explainability – Use Grad-CAM for better model interpretability.
- **Optimize for Telemedicine** – Adapt for mobile and remote healthcare use.
- **Clinical Validation** – Conduct real-world trials to assess performance.
- **Hybrid Models** – Combine clinical metadata with image analysis for better accuracy.
- **Privacy-Preserving AI** – Use federated learning for secure model training.

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THANK YOU!